

```
In [1]: import numpy as np
```

```
In [2]: np.__version__
```

```
Out[2]: '2.3.5'
```

```
In [3]: import sys  
sys.version
```

```
Out[3]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929 6  
4 bit (AMD64)]'
```

Creating Array

```
In [7]: my_list = [1,2,3,4,5,6,7,8,9]  
my_list
```

```
Out[7]: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
In [8]: type(my_list)
```

```
Out[8]: list
```

```
In [9]: a = np.array(my_list)
```

```
In [10]: a
```

```
Out[10]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [11]: type(a)
```

```
Out[11]: numpy.ndarray
```

```
In [12]: type(my_list)
```

```
Out[12]: list
```

important Function

1) arange

```
In [14]: np.arange(15)
```

```
Out[14]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
In [15]: np.arange(20)
```

```
Out[15]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19])
```

```
In [16]: np.arange(10,20)
```

```
Out[16]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [72]: np.arange(10,30,4)
```

```
Out[72]: array([10, 14, 18, 22, 26])
```

```
In [18]: np.arange(-20,-1)
```

```
Out[18]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
                -7,  -6,  -5,  -4,  -3,  -2])
```

```
In [19]: np.arange(30,10) # first Argument always be less than second argument
```

```
Out[19]: array([], dtype=int64)
```

```
In [20]: np.arange()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[20], line 1
----> 1 np.arange()

TypeError: arange() requires stop to be specified.
```

```
In [21]: np.arange(10,50,5)
```

```
Out[21]: array([10, 15, 20, 25, 30, 35, 40, 45])
```

```
In [22]: np.arange(10,20,5,2)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[22], line 1
----> 1 np.arange(10,20,5,2)

TypeError: Cannot interpret '2' as a data type
```

2) Zeros

```
In [25]: np.zeros(4) # by default gives float values
```

```
Out[25]: array([0., 0., 0., 0.])
```

```
In [26]: np.zeros(5,dtype = int) # convert float into the int
```

```
Out[26]: array([0, 0, 0, 0, 0])
```

```
In [27]: np.zeros((2,2),dtype = int )
```

```
Out[27]: array([[0, 0],
               [0, 0]])
```

```
In [29]: np.zeros((4,4),dtype = int)
```

```
Out[29]: array([[0, 0, 0, 0],
               [0, 0, 0, 0],
               [0, 0, 0, 0],
               [0, 0, 0, 0]])
```

```
In [30]: zero = np.zeros((2,2),dtype=int)
print(zero)
print('###')
print(type(zero))
```

```
[[0 0]
 [0 0]]
###
<class 'numpy.ndarray'>
```

```
In [33]: np.zeros((2,4)) # by default gives the float value
```

```
Out[33]: array([[0., 0., 0., 0.],
               [0., 0., 0., 0.]])
```

```
In [35]: np.zeros((3,3))
```

```
Out[35]: array([[0., 0., 0.],
               [0., 0., 0.],
               [0., 0., 0.]])
```

```
In [36]: np.zeros((4,5))
```

```
Out[36]: array([[0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0.]])
```

```
In [37]: np.zeros((4,5),dtype =int)
```

```
Out[37]: array([[0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0]])
```

```
In [40]: n1 = (9,10)
print(np.zeros((n1),dtype=int))
```

```
[[0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0]]
```

```
In [41]: n2 = (7,7)
         print(np.zeros((n2),dtype=int))
```

```
[[0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]]
```

```
In [42]: np.zeros((n2))
```

```
Out[42]: array([[0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0.]])
```

```
In [43]: n2
```

```
Out[43]: (7, 7)
```

```
In [45]: n3 = (20,20)
         print(np.zeros((n3)))
```

```
In [47]: print(np.zeros((n3),dtype = int ))
```

```
In [56]: print(np.zeros((3,4,2)))
```

```
[[[0. 0.]
  [0. 0.]
  [0. 0.]
  [0. 0.]]
```

```
[[0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]
```

```
[[0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]]
```

In [57]: `print(np.zeros((3,4,2,5)))`

```
[[[[0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]]
```

```
[[[0. 0. 0. 0. 0.]
  [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]]
```

```
[[[0. 0. 0. 0. 0.]
  [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]]
```

In [58]: `print(np.zeros((3,4,2,5,6)))`

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
```

```
[0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0.]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]]
```

```
[[[[0. 0. 0. 0. 0. 0.]
    [0. 0. 0. 0. 0. 0.]
    [0. 0. 0. 0. 0. 0.]
    [0. 0. 0. 0. 0. 0.]
    [0. 0. 0. 0. 0. 0.]]]]]
```



```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
```

```
[[[0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]
   [0. 0. 0. 0. 0. 0.]]]
```

```
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]]]
```

3) ones

```
In [48]: np.ones(3) # by default gives to the float value
```

```
Out[48]: array([1., 1., 1.])
```

```
In [50]: np.ones((3),dtype=int) # change the data type into the int
```

```
Out[50]: array([1, 1, 1])
```

```
In [51]: np.ones((3,2))
```

```
Out[51]: array([[1., 1.],
                [1., 1.],
                [1., 1.]])
```

```
In [52]: np.ones((3,10),dtype=int)
```

```
Out[52]: array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
```

```
In [55]: np.ones((3,4,2))
```

```
Out[55]: array([[[1., 1.],
                  [1., 1.],
                  [1., 1.],
                  [1., 1.]],

                [[1., 1.],
                  [1., 1.],
                  [1., 1.],
                  [1., 1.]],

                [[1., 1.],
                  [1., 1.],
                  [1., 1.],
                  [1., 1.]])
```

```
In [60]: np.ones((3,4,2,5),dtype= int)
```

```
Out[60]: array([[[[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]]],

              [[[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]]],

              [[[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]],

                [[1, 1, 1, 1, 1],
                  [1, 1, 1, 1, 1]]]])
```

```
In [61]: np.twos((2,4))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[61], line 1
----> 1 np.twos((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

```
In [62]: np.threes((4,5))
```

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[62], line 1
----> 1 np.threes((4,5))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'threes'

```

- Error Happens because the two & three is not the function

```
In [63]: range(0,5)
```

```
Out[63]: range(0, 5)
```

4) Random.rand (float) / Random.randint (int)

```
In [65]: np.random.rand(10)
```

```
Out[65]: array([0.79615068, 0.81022867, 0.98514013, 0.23858003, 0.49658383,
                0.58104113, 0.74479759, 0.45818076, 0.73332283, 0.97764135])
```

```
In [66]: np.random.randint(7)
```

```
Out[66]: 1
```

```
In [67]: np.random.randint(7)
```

```
Out[67]: 0
```

```
In [68]: np.random.randint(7)
```

```
Out[68]: 4
```

```
In [69]: np.random.rand(3,8)
```

```
Out[69]: array([[0.47204945, 0.13924936, 0.84714747, 0.11983055, 0.76189166,
                0.31401485, 0.41866534, 0.29545829],
                [0.037719 , 0.53617774, 0.1713295 , 0.75205999, 0.24789849,
                0.53138954, 0.13128925, 0.5174833 ],
                [0.15712678, 0.0278127 , 0.95543573, 0.15997598, 0.12764057,
                0.29643057, 0.10710832, 0.25340301]])
```

```
In [70]: np.random.randint(3,8)
```

```
Out[70]: 4
```

```
In [71]: np.random.randint(10,20,2)
```

```
Out[71]: array([16, 10], dtype=int32)
```

```
In [73]: np.random.randint(1,100,10)
```

```
Out[73]: array([24,  2, 94, 98, 92, 46, 78, 45,  9, 34], dtype=int32)
```

- first argument shows starting
- second argument show ending
- third argument show how random int you want

```
In [74]: np.random.randint(2,100,4,5)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[74], line 1
----> 1 np.random.randint(2,100,4,5)

File numpy/random/mtrand.pyx:775, in numpy.random.mtrand.RandomState.randint()

TypeError: Cannot interpret '5' as a data type
```

- above 3 argument this function shows error

```
In [78]: np.random.randint(1,100,(10,7))
```

```
Out[78]: array([[25, 59, 39,  7, 75,  2, 36],
 [15, 62, 91, 98, 62,  4,  8],
 [61, 49, 65,  7, 58, 55, 28],
 [53, 24, 54, 65, 97, 94,  5],
 [56, 56, 39, 69, 68, 41, 71],
 [65, 12, 78, 58, 22, 49, 64],
 [36, 61, 18, 41, 80, 76, 84],
 [22, 41, 34, 34, 57, 22, 50],
 [46, 37, 53,  1, 97, 83,  9],
 [22, 77, 33, 64, 26, 70, 67]], dtype=int32)
```

```
In [79]: np.random.randint(2,50,(5,5))
```

```
Out[79]: array([[46,  2, 12, 33, 46],
 [12,  5, 33, 42, 14],
 [43, 49, 45, 35, 29],
 [35, 15, 22, 19, 42],
 [32, 17, 29, 44, 48]], dtype=int32)
```

```
In [80]: np.random.randint(2,50,(5,5,2))
```

```
Out[80]: array([[[40, 12],
                 [16,  9],
                 [37, 39],
                 [ 9, 43],
                 [26, 41]],

                [[ 5, 34],
                 [ 2, 30],
                 [36, 34],
                 [29, 36],
                 [13, 21]],

                [[40, 25],
                 [27, 12],
                 [31, 33],
                 [32, 13],
                 [25, 32]],

                [[16, 42],
                 [23, 35],
                 [15, 35],
                 [ 5, 17],
                 [39,  5]],

                [[12, 20],
                 [43,  5],
                 [15, 14],
                 [16, 33],
                 [27, 29]]], dtype=int32)
```

```
In [89]: b = np.random.randint(1,100,(2,3,6,7))
         b
```

```
Out[89]: array([[[[34,  2, 82, 33, 34, 12, 70],
                  [71, 50, 61, 88, 56, 98, 59],
                  [73, 80, 92, 43, 70, 97, 73],
                  [ 6, 11, 36, 89, 41, 65, 15],
                  [47, 10, 23, 20, 64, 77, 89],
                  [85, 95, 54, 49, 12, 83, 85]],

                [[ 7, 34, 39, 29, 62, 72,  2],
                  [15, 71, 63, 57,  6, 55, 46],
                  [27,  2, 63, 25, 69, 39, 57],
                  [92, 64, 81, 45, 78, 35, 69],
                  [28, 87, 90, 25, 26, 86, 22],
                  [ 1, 73, 25, 47, 90, 76,  7]],

                [[45, 64, 27, 59, 38, 78, 93],
                  [49, 57, 17, 67, 47, 67, 53],
                  [88, 29, 59, 84, 50, 18, 14],
                  [64, 40, 25, 96, 84, 37, 17],
                  [79, 12, 98, 83, 98, 62, 11],
                  [41, 82, 51, 86, 72, 23, 87]]],

            [[[88, 86, 73, 15, 89, 47, 42],
              [56, 50, 76, 72, 60, 66, 58],
              [51, 41, 62, 58, 60, 82, 76],
              [33, 69, 71, 76, 52, 29, 96],
              [ 9, 78, 90, 51, 78, 16,  8],
              [25, 29, 97, 93, 98, 66, 50]],

            [[ 9, 30, 39, 30, 37, 51, 11],
              [94, 62, 82, 63,  2, 24, 54],
              [48, 24, 56, 33, 72, 12, 23],
              [17,  5, 43, 45, 84, 21, 70],
              [38, 11, 92, 87, 78, 31, 51],
              [98, 50,  3, 68, 99, 64, 79]],

            [[95, 49, 50, 61, 87, 57, 34],
              [35, 99, 52,  2,  6, 12,  9],
              [31, 72, 14, 85,  8, 25, 79],
              [64, 18, 68, 15, 44, 90, 79],
              [63, 60, 99,  3, 14, 86, 38],
              [63, 97, 35, 22, 85, 46,  4]]], dtype=int32)
```

- First argument show start
- second argument show end
- sixth argument show coloumn
- fifth coloume shows row
- fourth is the matrix of fifth row * sixth row
- three is mix of the privious all matrix in one matrix & create the big matrix

```
In [90]: type(b)
```

```
Out[90]: numpy.ndarray
```

```
In [93]: c = np.random.randint(10,20,(5,4))  
c
```

```
Out[93]: array([[13, 16, 18, 14],  
               [17, 10, 14, 10],  
               [14, 18, 18, 17],  
               [18, 10, 11, 18],  
               [19, 19, 15, 14]], dtype=int32)
```

```
In [79]: np.arange(1,13).reshape(3,4)
```

```
Out[79]: array([[ 1,  2,  3,  4],  
               [ 5,  6,  7,  8],  
               [ 9, 10, 11, 12]])
```

```
In [80]: np.arange(1,13).reshape(1,12)
```

```
Out[80]: array([[ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12]])
```

```
In [81]: np.arange(1,13).reshape(12,1)
```

```
Out[81]: array([[ 1],  
               [ 2],  
               [ 3],  
               [ 4],  
               [ 5],  
               [ 6],  
               [ 7],  
               [ 8],  
               [ 9],  
               [10],  
               [11],  
               [12]])
```

slicing

```
In [94]: c[:]
```

```
Out[94]: array([[13, 16, 18, 14],  
               [17, 10, 14, 10],  
               [14, 18, 18, 17],  
               [18, 10, 11, 18],  
               [19, 19, 15, 14]], dtype=int32)
```

```
In [96]: c[3:2] # this show empty because the starting argument is greter than ending
```

```
Out[96]: array([], shape=(0, 4), dtype=int32)
```

```
In [98]: c[1:3]
```



```
Out[98]: array([[17, 10, 14, 10],
               [14, 18, 18, 17]], dtype=int32)
```

```
In [101... c[1,3]
```

```
Out[101... np.int32(10)
```

```
In [4]: import numpy as np
d = np.random.randint(10,20,(4,5))
d
```

```
Out[4]: array([[10, 13, 11, 12, 13],
               [10, 16, 17, 15, 15],
               [13, 11, 12, 14, 17],
               [19, 13, 18, 15, 16]], dtype=int32)
```

```
In [5]: c = np.random.randint(10,20,(5,4))
c
```

```
Out[5]: array([[16, 10, 13, 19],
               [16, 19, 10, 14],
               [12, 10, 17, 13],
               [16, 10, 16, 18],
               [13, 15, 10, 19]], dtype=int32)
```

```
In [7]: c[1:3]
```

```
Out[7]: array([[16, 19, 10, 14],
               [12, 10, 17, 13]], dtype=int32)
```

```
In [8]: c[1,2]
```

```
Out[8]: np.int32(10)
```

```
In [10]: c[:]
```

```
Out[10]: array([[16, 10, 13, 19],
                [16, 19, 10, 14],
                [12, 10, 17, 13],
                [16, 10, 16, 18],
                [13, 15, 10, 19]], dtype=int32)
```

```
In [11]: c[-2,1]
```

```
Out[11]: np.int32(10)
```

```
In [12]: c[1,3]
```

```
Out[12]: np.int32(14)
```

```
In [13]: c[2:3]
```

```
Out[13]: array([[12, 10, 17, 13]], dtype=int32)
```

```
In [14]: c[0:-3]
```

```
Out[14]: array([[16, 10, 13, 19],  
               [16, 19, 10, 14]], dtype=int32)
```

```
In [15]: c[-3,-1]
```

```
Out[15]: np.int32(13)
```

Operation

```
In [17]: a = np.random.randint(10,20,10)  
a
```

```
Out[17]: array([10, 16, 15, 15, 12, 15, 16, 12, 13, 16], dtype=int32)
```

```
In [18]: id(a)
```

```
Out[18]: 1370593892304
```

```
In [22]: my_arr= np.array([1,2,3,4,5,6,7,8,9])  
my_arr
```

```
Out[22]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [24]: my_arr[0:6]
```

```
Out[24]: array([1, 2, 3, 4, 5, 6])
```

```
In [26]: my_arr2=np.random.randint(20,100,(10,10))
```

```
In [27]: my_arr2
```

```
Out[27]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],  
               [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],  
               [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],  
               [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],  
               [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],  
               [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],  
               [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],  
               [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],  
               [47, 79, 73, 47, 95, 58, 60, 35, 73, 37],  
               [45, 52, 49, 47, 79, 41, 25, 81, 39, 71]], dtype=int32)
```

```
In [28]: my_arr[:]
```

```
Out[28]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [29]: my_arr[:4]
```

```
Out[29]: array([1, 2, 3, 4])
```

```
In [30]: my_arr[:]
```

```
Out[30]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [31]: my_arr[0:6]
```

```
Out[31]: array([1, 2, 3, 4, 5, 6])
```

```
In [32]: my_arr2
```

```
Out[32]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],
                [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],
                [47, 79, 73, 47, 95, 58, 60, 35, 73, 37],
                [45, 52, 49, 47, 79, 41, 25, 81, 39, 71]], dtype=int32)
```

```
In [34]: my_arr2[2,7]
```

```
Out[34]: np.int32(31)
```

```
In [35]: my_arr2[-5,5]
```

```
Out[35]: np.int32(42)
```

```
In [37]: my_arr2[-2,3]
```

```
Out[37]: np.int32(47)
```

```
In [39]: my_arr2[::-1]
```

```
Out[39]: array([[45, 52, 49, 47, 79, 41, 25, 81, 39, 71],
                [47, 79, 73, 47, 95, 58, 60, 35, 73, 37],
                [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [25, 50, 74, 20, 69, 43, 40, 64, 52, 96]], dtype=int32)
```

```
In [40]: my_arr2[::-2]
```

```
Out[40]: array([[45, 52, 49, 47, 79, 41, 25, 81, 39, 71],
                [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47]], dtype=int32)
```

```
In [42]: my_arr2[::-3]
```

```
Out[42]: array([[45, 52, 49, 47, 79, 41, 25, 81, 39, 71],
               [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],
               [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
               [25, 50, 74, 20, 69, 43, 40, 64, 52, 96]], dtype=int32)
```

```
In [43]: my_arr2[:-3]
```

```
Out[43]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
               [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
               [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
               [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
               [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
               [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
               [96, 48, 78, 40, 84, 64, 93, 24, 64, 60]], dtype=int32)
```

```
In [46]: my_arr
```

```
Out[46]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

5) Min

```
In [47]: my_arr
```

```
Out[47]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [49]: my_arr.min()
```

```
Out[49]: np.int64(1)
```

```
In [50]: my_arr2.min()
```

```
Out[50]: np.int32(20)
```

```
In [52]: my_arr3 = np.array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                             [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                             [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                             [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                             [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                             [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                             [96, 48, 78, 40, 84, 64, 93, 24, 64, 60]])
my_arr3
```

```
Out[52]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
               [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
               [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
               [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
               [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
               [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
               [96, 48, 78, 40, 84, 64, 93, 24, 64, 60]])
```

```
In [53]: my_arr.min()
```

```
Out[53]: np.int64(1)
```

6) max

```
In [55]: my_arr
```

```
Out[55]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [56]: my_arr.max()
```

```
Out[56]: np.int64(9)
```

```
In [57]: my_arr2
```

```
Out[57]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],
                [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],
                [47, 79, 73, 47, 95, 58, 60, 35, 73, 37],
                [45, 52, 49, 47, 79, 41, 25, 81, 39, 71]], dtype=int32)
```

```
In [59]: my_arr2.max()
```

```
Out[59]: np.int32(99)
```

```
In [60]: my_arr3
```

```
Out[60]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60]])
```

```
In [61]: my_arr3.max()
```

```
Out[61]: np.int64(99)
```

7) Mean

```
In [62]: my_arr
```

```
Out[62]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [63]: my_arr.mean()
```

```
Out[63]: np.float64(5.0)
```

```
In [64]: my_arr2
```

```
Out[64]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60],
                [54, 84, 22, 67, 27, 39, 33, 66, 26, 40],
                [47, 79, 73, 47, 95, 58, 60, 35, 73, 37],
                [45, 52, 49, 47, 79, 41, 25, 81, 39, 71]], dtype=int32)
```

```
In [65]: my_arr2.mean()
```

```
Out[65]: np.float64(57.61)
```

```
In [66]: my_arr3
```

```
Out[66]: array([[25, 50, 74, 20, 69, 43, 40, 64, 52, 96],
                [83, 62, 88, 92, 31, 66, 66, 40, 84, 47],
                [38, 91, 95, 77, 50, 57, 48, 31, 28, 91],
                [20, 42, 21, 43, 36, 75, 77, 46, 75, 62],
                [68, 81, 86, 57, 34, 40, 24, 79, 57, 51],
                [90, 74, 99, 51, 79, 42, 38, 45, 88, 41],
                [96, 48, 78, 40, 84, 64, 93, 24, 64, 60]])
```

```
In [67]: my_arr3.mean()
```

```
Out[67]: np.float64(59.57142857142857)
```

```
In [ ]: median
```

8) Median

```
In [68]: my_arr
```

```
Out[68]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [77]: import numpy as np
ar= array([1, 2, 3, 4, 5, 6, 7, 8, 9])
median(ar)
```

```
Out[77]: np.float64(5.0)
```

```
In [76]: from numpy import *  
a = array([1,2,3,4,9])  
median(a)
```

```
Out[76]: np.float64(3.0)
```

```
In [ ]:
```